

Element

An **element** is a substance that cannot be broken down into simpler substances by ordinary chemical means. Each element is defined by its **atomic number**, Z , the number of protons in the nucleus of its atoms — change the number of protons and you have a different element entirely. As of the current periodic table, 118 elements are known, arranged by increasing atomic number and organized so that elements with similar chemical behavior fall into the same vertical column, or group.

Worked example. Consider carbon ($Z = 6$) versus nitrogen ($Z = 7$). Both are common in biological molecules, but a carbon atom has 6 protons while a nitrogen atom has 7. Because the number of protons differs, no chemical reaction can convert carbon into nitrogen — that transformation requires nuclear physics (fusion, fission, or radioactive decay), not chemistry. This is the practical test for "elementhood": if a process only rearranges electrons and bonds, the elements present before and after are identical.

Problem-solving application. Suppose you are given an unknown pure substance and told it decomposes under heat into two separate gases with different densities. Is it an element? No — by definition, an element cannot decompose into anything simpler. The observation that it split into two distinct substances tells you it was a compound, and the two gases produced are candidates for being elements (or could themselves be further decomposable compounds, which you would test the same way). This decomposition test is exactly how many elements, such as hydrogen and oxygen from water, were historically identified.

A related problem-solving skill is reading atomic number from context: given that a particle has 11 protons, you can identify it as sodium no matter what else is true about it — whether it has gained or lost electrons, or how many protons pair with it in the nucleus — because protons, and protons alone, define elemental identity. This is why sodium always sits in the same position on the periodic table: its identity as an element is fixed by Z , even when its charge or exact mass varies from one sample to another.

